EFY Note
The source code of this project is available for free download at *source.efymag.com*

```
do
{
        Key=0;
        R1=R2=R3=R4=0;
        while(Key==0);
}while(Key!='#');
```

Fig. 7: Pseudo-code source listing for detecting a keypress in main application

5. Inside the ISR, the actual checking or identifying the specific row's key, which is responsible for the key press, is detected and the confirmed key is kept in the shared resource variable 'Key' and the ISR ends by forcing it to return. The suspended application will now be ready by context retrieval using stack's POP operations.

6. The pseudo-code shown in Fig. 7 is proposed for detecting a valid key pressed in main code for further application development.

7. Another important inference is that there will be no key debouncing problem, unlike in polled-loop keypad. As a result, the intentional delay for every consecutive key press is not necessary.

By considering this approach of interrupt-based keypad driver, one can implement and achieve the real-time response for UI without porting the RTOS into the MCU. This is because the ISR will run irrespective of what the application is doing, forcing the CPU's attention. Therefore, as a result if the design is done with this recommended UI, apart from achieving the real-time response, the memory footprint (both Flash and RAM) is also greatly reduced.

After downloading the source programs from EFY website, using circuit shown in Fig.8 along with Project2, the polled-loop application is tested with any 4-digit input as user ID and '1982' as password. The ISR-locked keypad driver is tested using Fig.4 and Project3. Finally, the same application with LCD is tested using Fig. 9 and Project4. EFY

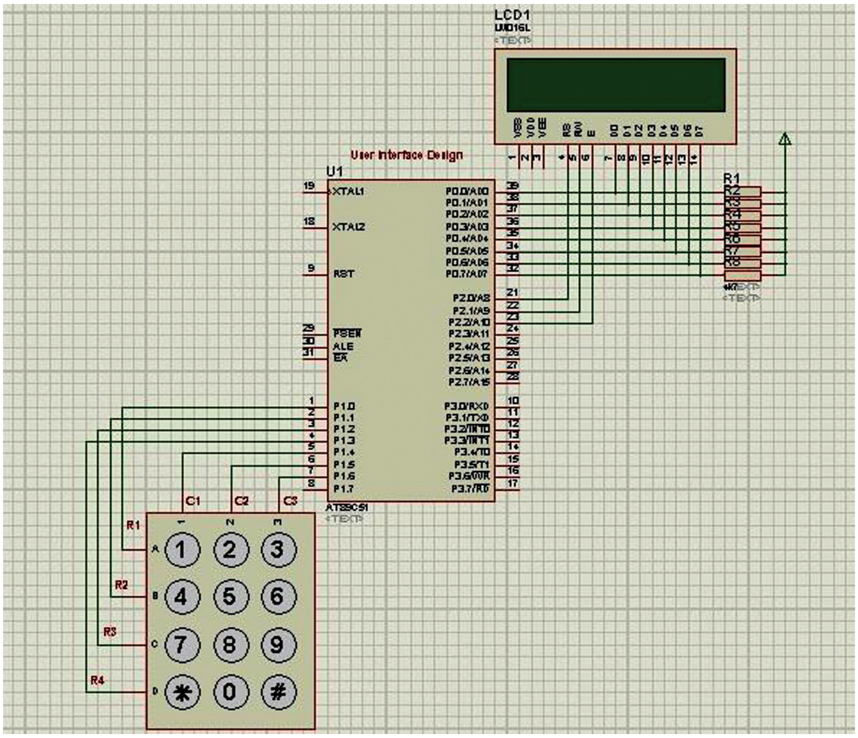

Fig. 8: User interface design with interrupt based matrix keypad and LCD (8-bit)

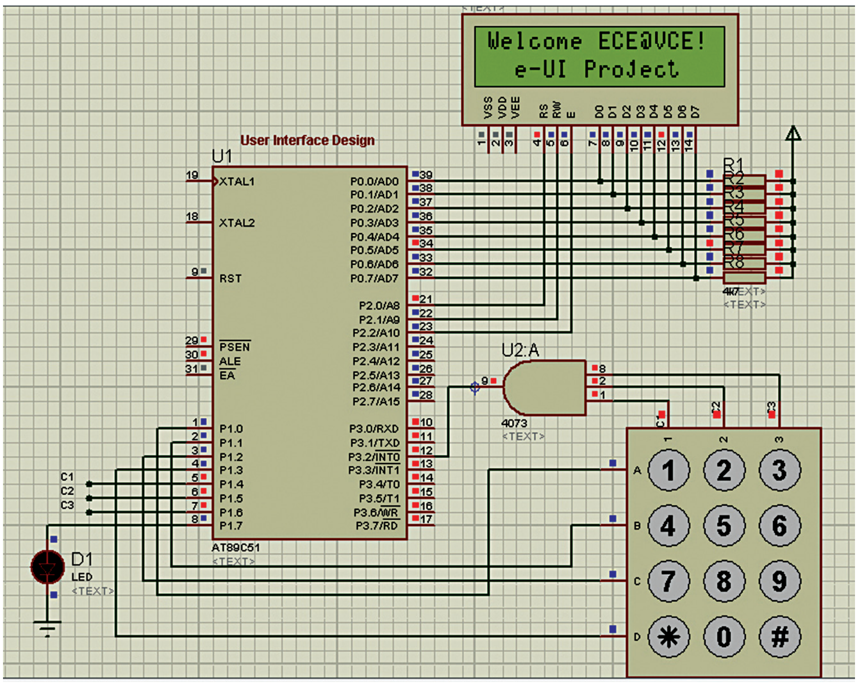

Fig. 9: User interface design with interrupt based matrix keypad using LCD and LED

N. Abid Ali Khan is Assistant Professor in the Department of ECE at Vasavi College of Engineering (Autonomous), Hyderabad. He has been working on power-efficient embedded networking architectures and is passionate about microcontrollers, ARM, RTOS, embedded systems, and programming